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Feature Analysis for Effective Tropical Cyclone Forecasting: Evaluating the Relevance of Data Inputs

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ABSTRACT

Accurate forecasting of tropical cyclones is crucial for minimizing the impact of these devastating natural phenomena. This study focuses on feature analysis to enhance the effectiveness of tropical cyclone forecasting models. By evaluating the relevance of various data inputs, including but not limited to latitude, longitude, wind speed, direction, and atmospheric pressure, this research aims to identify the most significant predictors for precise cyclone trajectory predictions. Through rigorous analysis and validation, this research aims to determine which data inputs are most critical for enhancing the reliability and accuracy of tropical cyclone forecasting. The findings will contribute valuable insights into feature selection strategies, aiding in the development of more effective and robust forecasting systems for disaster management and emergency response.

***Keywords**: Forecasting, Cyclone Landfall Forecasting,*

1. INTRODUCTION

Tropical cyclones, also known as typhoons or hurricanes, are extremely harsh weather phenomenon. Between 1998 and 2017, storms, including those of tropical cyclones and hurricanes, came second only to earthquakes causing fatalities. These killed 233,000 people and during the same period injured, made homeless, displaced, or evacuated about 726 million people worldwide during the emergency phase of the disaster. Over the last 30 years, the number of people living on coasts exposed to cyclones rose by 192%, raising mortality and morbidity in the case of a tropical cyclone [1].

The geographic location of the Philippines makes it highly vulnerable to tropical cyclones, which cause extensive damage and loss of life, particularly in regions like the Cagayan Valley [2]. Understanding and predicting the path of tropical storms could be of great help in enhancing disaster preparedness, risk mitigation efforts, and emergency response strategies in vulnerable coastal regions.

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